POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 3032 | Semester III | Theory | 4 Credits

Electrical Machines I

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 3032 as **Electrical Machines I** in Semester III for Electrical & Electronics Engineering. The official SITTTR source was unavailable during this cycle. With user approval, explanatory coverage is based on reliable public Electrical Machines I curricula and is not represented as recovered official Course 3032 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment entries are provisional handbook placeholders because the official SITTTR source could not be accessed. Study modules cover magnetic circuits, DC machines and transformers according to the cited public curricula; confirm exact official coverage when the SITTTR detail page returns.

Verified source note: The official SITTTR programme/detail source remained inaccessible during this automated cycle. The public course curricula used for explanations are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electrical Machines I introduces magnetic energy conversion, DC generators and motors, and single-/three-phase transformers. It connects construction, phasors and equivalent circuits to performance, losses, efficiency, tests and safe operation.

Malayalam support: □ handbook □□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□.

Prerequisite foundation

Magnetic circuits, AC/DC circuit theory, phasors and basic power concepts.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain magnetic circuits and electromechanical energy conversion.
2. Describe DC-generator construction, operation and characteristics.
3. Describe DC-motor operation, control, performance and testing.
4. Explain single- and three-phase transformer operation, tests, connections and applications.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain magnetic circuits and basic electromechanical energy conversion.	Understand
CO2	Explain construction, excitation, characteristics and operation of DC generators.	Apply
CO3	Explain DC-motor performance, starting, speed control, losses and testing.	Apply
CO4	Explain transformer operation, testing, efficiency, regulation and common three-phase connections.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Magnetic Circuits and Energy Conversion	MMF, flux, reluctance, inductance, magnetic fields, force/torque and energy conversion.	Approx. 12 hours	CO1	Module 1
2	DC Generators	Construction, EMF equation, excitation, armature reaction, commutation, characteristics and parallel operation.	Approx. 15 hours	CO2	Module 2
3	DC Motors and Testing	Back EMF, torque, shunt/series/compound motors, characteristics, starting, speed control, losses, efficiency and tests.	Approx. 15 hours	CO3	Module 3
4	Transformers	Single-phase principle/equivalent circuit/tests; efficiency/regulation; three-phase connections, auto transformers, tap changers and cooling.	Approx. 18 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Magnetic Circuits and Energy Conversion

MMF, flux, reluctance, inductance, magnetic fields, force/torque and energy conversion.

CO1 · Approx. 12 hours

DC Generators

Construction, EMF equation, excitation, armature reaction, commutation, characteristics and parallel operation.

CO2 · Approx. 15 hours

DC Motors and Testing

Back EMF, torque, shunt/series/compound motors, characteristics, starting, speed control, losses, efficiency and tests.

CO3 · Approx. 15 hours

Transformers

Single-phase principle/equivalent circuit/tests; efficiency/regulation; three-phase connections, auto transformers, tap changers and cooling.

CO4 · Approx. 18 hours

MODULE 1

Magnetic Circuits and Energy Conversion

MMF, flux, reluctance, inductance, magnetic fields, force/torque and energy conversion.

Approx. 12 hours | CO1 | Understand · Apply

Learning goals

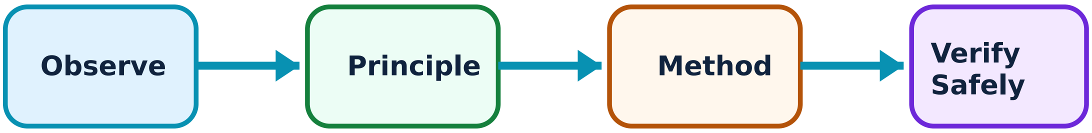
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Magnetic Circuits and Energy Conversion: identify the system, state its principle, follow the correct method and verify the result safely.

1. Magnetic-circuit quantities−

Magnetomotive force drives flux through a magnetic path against reluctance. Permeability, air gaps and material properties influence flux and inductance in machine magnetic circuits.

Study and application method

Draw the flux path, list each section’s length/area/material, state assumptions and use a consistent magnetic-circuit analogy only as a guide.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw the flux path, list each section's length/area/material, state assumptions and use a consistent magnetic-circuit analogy only as a guide.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Air gaps can dominate reluctance; do not neglect them in a magnetic circuit.

2. Electromechanical energy conversion−

Electrical machines exchange electrical and mechanical energy through a magnetic field. Force and torque arise from field interaction and motion, with energy balance guiding interpretation.

Study and application method

Identify electrical input/output, mechanical input/output and stored-field energy; label direction of power transfer for motor or generator operation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify electrical input/output, mechanical input/output and stored-field energy; label direction of power transfer for motor or generator operation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Do not confuse energy conversion direction with physical direction of rotation alone.

3. Armature windings and losses−

DC armature windings and commutators support generated voltage and current collection. Eddy-current and related losses motivate laminated cores and correct material/design choices.

Study and application method

Use a labelled armature sketch, distinguish lap/wave winding purpose and state one loss-reduction measure.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a labelled armature sketch, distinguish lap/wave winding purpose and state one loss-reduction measure.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Loss mechanisms must be identified separately; not all losses vary in the same way with load or speed.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

DC Generators

Construction, EMF equation, excitation, armature reaction, commutation, characteristics and parallel operation.

Approx. 15 hours | CO2 | Understand · Apply

Learning goals

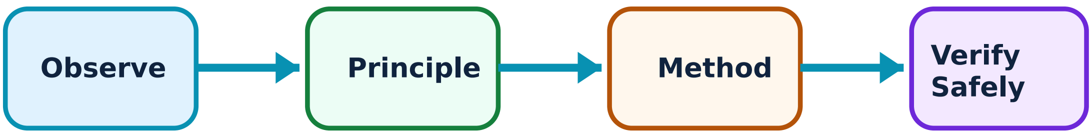
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for DC Generators: identify the system, state its principle, follow the correct method and verify the result safely.

1. Construction, principle and EMF generation−

A DC generator converts mechanical input to electrical output through electromagnetic induction, using field system, armature, commutator and brushes. Generated EMF depends on flux, speed and winding parameters.

Study and application method

Draw and label the machine, state energy conversion, then write the EMF relation only after identifying each symbol and operating condition.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw and label the machine, state energy conversion, then write the EMF relation only after identifying each symbol and operating condition.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Commutator action does not create energy; it provides the required output form/collection arrangement.

2. Excitation and characteristics−

Separately excited, shunt, series and compound generators differ in field connection and voltage behaviour. Open-circuit and load characteristics support selection and performance interpretation.

Study and application method

Identify field connection, sketch characteristic axes, state voltage-build-up condition and explain the effect of load using field/armature interaction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify field connection, sketch characteristic axes, state voltage-build-up condition and explain the effect of load using field/armature interaction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not use one generator characteristic for a different excitation connection.

3. Armature reaction, commutation and parallel operation−

Armature reaction distorts the main field and affects commutation. Interpoles/compensating methods improve commutation. Parallel operation requires correct conditions for stable load sharing.

Study and application method

Explain cause → effect → remedy, and use a connection/sequence diagram for parallel operation with safety checks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain cause → effect → remedy, and use a connection/sequence diagram for parallel operation with safety checks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Never connect machines in parallel without verified polarity, voltage and approved procedure.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

DC Motors and Testing

Back EMF, torque, shunt/series/compound motors, characteristics, starting, speed control, losses, efficiency and tests.

Approx. 15 hours | **CO3** | **Understand · Apply**

Learning goals

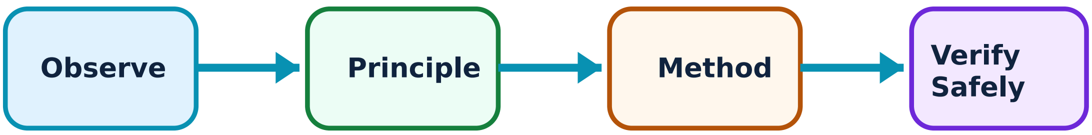
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for DC Motors and Testing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Motor principle, back EMF and torque−

A DC motor converts electrical input to mechanical output. Back EMF opposes supply as rotation develops, while torque arises from armature current and magnetic flux interaction.

Study and application method

Draw motor power flow, state back-EMF role and relate torque to flux/current qualitatively before using a formula.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw motor power flow, state back-EMF role and relate torque to flux/current qualitatively before using a formula.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: At starting, back EMF is low; starting current must be controlled by an approved starter.

2. Characteristics and speed control−

Shunt, series and compound motors have different speed–torque behaviour and applications. Speed control changes field, armature or supply conditions depending on machine type.

Study and application method

Compare characteristic shape, typical load and control method in a table; state operational limit and one safety check.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare characteristic shape, typical load and control method in a table; state operational limit and one safety check.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: A series motor should not be run without appropriate load because speed can become dangerously high.

3. Losses, efficiency and testing−

Copper, core, mechanical and stray losses reduce efficiency. Direct/indirect tests estimate performance and support safe machine evaluation.

Study and application method

Prepare a power-flow diagram, classify losses and choose a test with measured quantities, assumptions and required observation sheet.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a power-flow diagram, classify losses and choose a test with measured quantities, assumptions and required observation sheet.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Testing must use rated limits and safe guarding; do not treat a test setup as a casual live experiment.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Transformers

Single-phase principle/equivalent circuit/tests; efficiency/regulation; three-phase connections, auto transformers, tap changers and cooling.

Approx. 18 hours | **CO4** | **Understand · Apply**

Learning goals

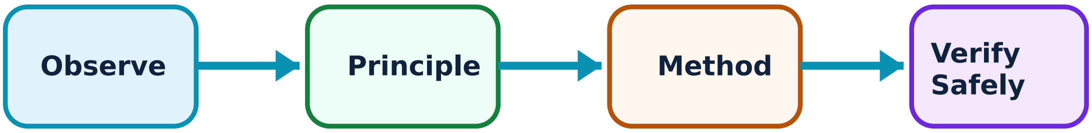
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Transformers: identify the system, state its principle, follow the correct method and verify the result safely.

1. Single-phase transformer operation−

A transformer transfers AC electrical energy between circuits through mutual flux. Turns ratio determines ideal voltage/current relation, while leakage, winding resistance and core loss affect practical operation.

Study and application method

Draw core/windings and flux path, state ratio, construct a phasor/equivalent-circuit explanation and distinguish no-load from loaded condition.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw core/windings and flux path, state ratio, construct a phasor/equivalent-circuit explanation and distinguish no-load from loaded condition.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: A transformer requires alternating flux; do not apply a DC supply to an ordinary transformer winding.

2. Tests, losses, efficiency and regulation−

Open-circuit and short-circuit tests estimate core/copper loss and equivalent parameters. Efficiency and voltage regulation describe useful performance under load and power factor.

Study and application method

State objective of each test, show safe connection, record measured quantities and use the intended calculation relation with clearly named losses.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State objective of each test, show safe connection, record measured quantities and use the intended calculation relation with clearly named losses.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Short-circuit testing uses reduced applied voltage; never apply full voltage to a shorted transformer.

3. Three-phase connections and special transformers−

Star/delta connections affect phase/line relationships, phase shift and application. Auto transformers, tap changers and cooling arrangements meet specific voltage-control and installation needs.

Study and application method

Draw connection and vector group at required level, compare voltage ratio/isolation/application and state one operational precaution.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw connection and vector group at required level, compare voltage ratio/isolation/application and state one operational precaution.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: An autotransformer does not provide the same galvanic isolation as a two-winding transformer.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Magnetic Circuits and Energy Conversion

Use the concept flow to revise observation, principle, method and verification.

Module 2: DC Generators

Use the concept flow to revise observation, principle, method and verification.

Module 3: DC Motors and Testing

Use the concept flow to revise observation, principle, method and verification.

Module 4: Transformers

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Draw a magnetic-circuit and energy-conversion diagram with symbols/units.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare DC generator excitation methods and characteristics using a table.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a DC-motor starting/speed-control safety checklist and a power-flow diagram.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw transformer test connections and compare one three-phase connection with an autotransformer application.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Magnetic Circuits and Energy Conversion

Explain Magnetic-circuit quantities and state one correct method, application or precaution.−

Magnetomotive force drives flux through a magnetic path against reluctance. Permeability, air gaps and material properties influence flux and inductance in machine magnetic circuits.

Answer framework: Draw the flux path, list each section's length/area/material, state assumptions and use a consistent magnetic-circuit analogy only as a guide.

Important point: Air gaps can dominate reluctance; do not neglect them in a magnetic circuit.

Explain Electromechanical energy conversion and state one correct method, application or precaution.−

Electrical machines exchange electrical and mechanical energy through a magnetic field. Force and torque arise from field interaction and motion, with energy balance guiding interpretation.

Answer framework: Identify electrical input/output, mechanical input/output and stored-field energy; label direction of power transfer for motor or generator operation.

Important point: Do not confuse energy conversion direction with physical direction of rotation alone.

Explain Armature windings and losses and state one correct method, application or precaution.−

DC armature windings and commutators support generated voltage and current collection. Eddy-current and related losses motivate laminated cores and correct material/design choices.

Answer framework: Use a labelled armature sketch, distinguish lap/wave winding purpose and state one loss-reduction measure.

Important point: Loss mechanisms must be identified separately; not all losses vary in the same way with load or speed.

Module 2 — DC Generators

Explain Construction, principle and EMF generation and state one correct method, application or precaution.−

A DC generator converts mechanical input to electrical output through electromagnetic induction, using field system, armature, commutator and brushes. Generated EMF depends on flux, speed and winding parameters.

Answer framework: Draw and label the machine, state energy conversion, then write the EMF relation only after identifying each symbol and operating condition.

Important point: Commutator action does not create energy; it provides the required output form/collection arrangement.

Explain Excitation and characteristics and state one correct method, application or precaution.−

Separately excited, shunt, series and compound generators differ in field connection and voltage behaviour. Open-circuit and load characteristics support selection and performance interpretation.

Answer framework: Identify field connection, sketch characteristic axes, state voltage-build-up condition and explain the effect of load using field/armature interaction.

Important point: Do not use one generator characteristic for a different excitation connection.

Explain Armature reaction, commutation and parallel operation and state one correct method, application or precaution.−

Armature reaction distorts the main field and affects commutation. Interpoles/compensating methods improve commutation. Parallel operation requires correct conditions for stable load sharing.

Answer framework: Explain cause → effect → remedy, and use a connection/sequence diagram for parallel operation with safety checks.

Important point: Never connect machines in parallel without verified polarity, voltage and approved procedure.

Module 3 — DC Motors and Testing

Explain Motor principle, back EMF and torque and state one correct method, application or precaution.−

A DC motor converts electrical input to mechanical output. Back EMF opposes supply as rotation develops, while torque arises from armature current and magnetic flux interaction.

Answer framework: Draw motor power flow, state back-EMF role and relate torque to flux/current qualitatively before using a formula.

Important point: At starting, back EMF is low; starting current must be controlled by an approved starter.

Explain Characteristics and speed control and state one correct method, application or precaution.−

Shunt, series and compound motors have different speed–torque behaviour and applications. Speed control changes field, armature or supply conditions depending on machine type.

Answer framework: Compare characteristic shape, typical load and control method in a table; state operational limit and one safety check.

Important point: A series motor should not be run without appropriate load because speed can become dangerously high.

Explain Losses, efficiency and testing and state one correct method, application or precaution.−

Copper, core, mechanical and stray losses reduce efficiency. Direct/indirect tests estimate performance and support safe machine evaluation.

Answer framework: Prepare a power-flow diagram, classify losses and choose a test with measured quantities, assumptions and required observation sheet.

Important point: Testing must use rated limits and safe guarding; do not treat a test setup as a casual live experiment.

Module 4 — Transformers

Explain Single-phase transformer operation and state one correct method, application or precaution.−

A transformer transfers AC electrical energy between circuits through mutual flux. Turns ratio determines ideal voltage/current relation, while leakage, winding resistance and core loss affect practical operation.

Answer framework: Draw core/windings and flux path, state ratio, construct a phasor/equivalent-circuit explanation and distinguish no-load from loaded condition.

Important point: A transformer requires alternating flux; do not apply a DC supply to an ordinary transformer winding.

Explain Tests, losses, efficiency and regulation and state one correct method, application or precaution.−

Open-circuit and short-circuit tests estimate core/copper loss and equivalent parameters. Efficiency and voltage regulation describe useful performance under load and power factor.

Answer framework: State objective of each test, show safe connection, record measured quantities and use the intended calculation relation with clearly named losses.

Important point: Short-circuit testing uses reduced applied voltage; never apply full voltage to a shorted transformer.

Explain Three-phase connections and special transformers and state one correct method, application or precaution.−

Star/delta connections affect phase/line relationships, phase shift and application. Auto transformers, tap changers and cooling arrangements meet specific voltage-control and installation needs.

Answer framework: Draw connection and vector group at required level, compare voltage ratio/isolation/application and state one operational precaution.

Important point: An autotransformer does not provide the same galvanic isolation as a two-winding transformer.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Magnetic Circuits and Energy Conversion.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: DC Generators.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: DC Motors and Testing.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Transformers.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: MMF, flux, reluctance, inductance, magnetic fields, force/torque and energy conversion..
2. Develop a complete answer or practical plan for: Construction, EMF equation, excitation, armature reaction, commutation, characteristics and parallel operation..
3. Develop a complete answer or practical plan for: Back EMF, torque, shunt/series/compound motors, characteristics, starting, speed control, losses, efficiency and tests..
4. Develop a complete answer or practical plan for: Single-phase principle/equivalent circuit/tests; efficiency/regulation; three-phase connections, auto transformers, tap changers and cooling..

LAST-MILE REVISION

Quick Revision

Module 1: Magnetic Circuits and Energy Conversion

MMF, flux, reluctance, inductance, magnetic fields, force/torque and energy conversion.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: DC Generators

Construction, EMF equation, excitation, armature reaction, commutation, characteristics and parallel operation.

- Match each topic to CO2.

- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: DC Motors and Testing

Back EMF, torque, shunt/series/compound motors, characteristics, starting, speed control, losses, efficiency and tests.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Transformers

Single-phase principle/equivalent circuit/tests; efficiency/regulation; three-phase connections, auto transformers, tap changers and cooling.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Magnetic-circuit quantities	Magnetomotive force drives flux through a magnetic path against reluctance.
Electromechanical energy conversion	Electrical machines exchange electrical and mechanical energy through a magnetic field.
Armature windings and losses	DC armature windings and commutators support generated voltage and current collection.
Construction, principle and EMF generation	A DC generator converts mechanical input to electrical output through electromagnetic induction, using field system, armature, commutator and brushes.
Excitation and characteristics	Separately excited, shunt, series and compound generators differ in field connection and voltage behaviour.
Armature reaction, commutation and parallel operation	Armature reaction distorts the main field and affects commutation.
Motor principle, back EMF and torque	A DC motor converts electrical input to mechanical output.
Characteristics and speed control	Shunt, series and compound motors have different speed–torque behaviour and applications.
Losses, efficiency and testing	Copper, core, mechanical and stray losses reduce efficiency.
Single-phase transformer operation	A transformer transfers AC electrical energy between circuits through mutual flux.
Tests, losses, efficiency and regulation	Open-circuit and short-circuit tests estimate core/copper loss and equivalent parameters.
Three-phase connections and special transformers	Star/delta connections affect phase/line relationships, phase shift and application.

LEARNING RESOURCES

References

Recommended electrical-machines references

1. P. S. Bimbhra, Electrical Machinery.
2. I. J. Nagrath and D. P. Kothari, Electric Machines.
3. A. E. Fitzgerald, C. Kingsley and S. Umans, Electric Machinery.

Approved public Electrical Machines I sources

- https://www.iare.ac.in/sites/default/files/Courses_description/EM-I%20R-18%20syllabus.pdf

- https://ee.cet.ac.in/program/btech/EE205%20DC%20Machines%20n%20transformers.pdf

The linked curricula support study explanations while the official Course 3032 detail page is unavailable; they do not replace an official detailed syllabus.